

CN Unit 4 Question Bank — Transport Layer (TCP/UDP)

Computer Networks (CN) · Unit 4 · Transport Layer (TCP/UDP) · 2025

Important and frequently asked questions, grouped by marks.

Part A — Short answer (2 marks)

1. Explain UDP with a neat diagram and a suitable example.
2. Compare TCP Connection Management with Quality of Service. When would you prefer one over the other?
3. Compare TCP Flow Control with TCP Connection Management. When would you prefer one over the other?
4. What is meant by TCP Congestion Control? Discuss its advantages and limitations.
5. Compare Port Numbers and Sockets with UDP. When would you prefer one over the other?
6. Illustrate Quality of Service with a worked example.

Part B — Long answer (10 marks)

1. Discuss the role of UDP in Transport Layer (TCP/UDP).
2. Define TCP Connection Management. Derive or justify the key properties involved.
3. Illustrate TCP Flow Control with a worked example.
4. Explain TCP Congestion Control with a neat diagram and a suitable example.
5. Illustrate Port Numbers and Sockets with a worked example.
6. Explain Quality of Service with a neat diagram and a suitable example.
7. What is meant by UDP? Discuss its advantages and limitations.
8. What is meant by TCP Connection Management? Discuss its advantages and limitations.
9. Illustrate TCP Flow Control with a worked example.
10. Write short notes on TCP Congestion Control.
11. Illustrate Port Numbers and Sockets with a worked example.
12. Compare Quality of Service with TCP Connection Management. When would you prefer one over the other?